QSCP Corporate Asset Integrity

Corporate Summary

No.	Status	Date
QSCP-CI-SUM-2014-02	Issued	14.07.2014

Corporate Integrity Summary - External Chloride SCC, Anlage 21

Incident Overview and Initial Findings

During activities related to a 2013 ZÜS deficiency, a nonconformity was raised under NR-AB-2014-03 regarding line NR-21-L-0412 in ANLAGE-21. The 1.4404 stainless steel line, which transports K-2101 bottoms to E-2104, was found with wet insulation and damaged cladding. Upon removal of the insulation, penetrant testing revealed surface indications, leading to the line being taken out of service. The spool has a design pressure of 10 bar(g) and a design temperature of 150 °C.

Metallographic and Root Cause Analysis

A report by Rhein-Lippe Werkstoffprüfung GmbH, RLW-PB-2014-0131, details the metallographic investigation of the removed section. The analysis confirmed transgranular branched cracking originating from the outer surface, which is characteristic of chloride-induced stress corrosion cracking. The root-cause analysis, NR-UA-2014-01, identified that rain water entered through the compromised cladding. This environmental factor, combined with a 2011 process change that increased operating temperatures to 95 °C, facilitated the external chloride SCC mechanism.

Corporate Requirements

This summary, prepared by Daniel Hartmann, serves to alert all QSCP sites to the risks of external SCC under insulation when cladding integrity is breached on austenitic stainless steel systems. Sites must ensure that insulation systems are maintained to prevent moisture ingress, particularly on lines where process temperatures have been increased.

Sign-off	Name	Function	Date	Signature
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